

Theorem 1 Let U and V be finite-dimensional vector subspaces over $\mathbf{F}$. Then

$$\dim(U + V) = \dim U + \dim V - \dim(U \cap V).$$

Proof. Let $W = U \cap V$ and set $\dim W = p$, $\dim U = m + p$, and $\dim V = n + p$. Pick a basis $B_W = \{w_1, \dots, w_p\}$ for W and extend it to bases

$$B_U = \{w_1, \dots, w_p, u_1, \dots, u_m\} \quad \text{for } U, \quad B_V = \{w_1, \dots, w_p, v_1, \dots, v_n\} \quad \text{for } V.$$

We claim that

$$B_{U+V} := \{w_1, \dots, w_p, u_1, \dots, u_m, v_1, \dots, v_n\}$$

is a basis for $U + V$.

Spanning. Let $x \in U + V$. Then $x = u + v$ for some $u \in U$ and $v \in V$, so

$$x = \sum_{i=1}^p b_i w_i + \sum_{i=1}^m b_{p+i} u_i + \sum_{i=1}^p c_i w_i + \sum_{i=1}^n c_{p+i} v_i.$$

Collecting the w_i terms (set $a_i = b_i + c_i$ for $i = 1, \dots, p$),

$$x = \sum_{i=1}^p a_i w_i + \sum_{i=1}^m b_{p+i} u_i + \sum_{i=1}^n c_{p+i} v_i \in \text{span}(B_{U+V}),$$

so $\text{span}(B_{U+V}) = U + V$.

Linear independence. Suppose

$$\sum_{i=1}^p a_i w_i + \sum_{i=1}^m b_i u_i + \sum_{i=1}^n c_i v_i = 0.$$

Define

$$w := \sum_{i=1}^p a_i w_i + \sum_{i=1}^m b_i u_i = - \sum_{i=1}^n c_i v_i.$$

The left sum lies in U , the right in V , so $w \in W$. Write $w = \sum_{i=1}^p d_i w_i$ using the basis $B_W \subseteq B_V$. Then

$$0 = w + \sum_{i=1}^n c_i v_i = \sum_{i=1}^p d_i w_i + \sum_{i=1}^n c_i v_i,$$

which is a linear combination of the independent set B_V . Hence

$$d_1 = \dots = d_p = 0 \quad \text{and} \quad c_1 = \dots = c_n = 0.$$

Therefore $w = 0$, and the earlier expression becomes

$$0 = \sum_{i=1}^p a_i w_i + \sum_{i=1}^m b_i u_i,$$

so independence of B_U forces $a_1 = \dots = a_p = b_1 = \dots = b_m = 0$. Hence B_{U+V} is linearly independent.

Therefore B_{U+V} is a basis, so $\dim(U + V) = p + m + n = (m + p) + (n + p) - p = \dim U + \dim V - \dim W$. $\square$